# How the three ensembles come to agree: an exact study of a two-dimensional quantum lattice gas

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Exact diagonalization of the hard-core Bose–Hubbard model

**Abstract.** Statistical mechanics asserts that the microcanonical, canonical, and grand-canonical ensembles agree in the thermodynamic limit. For a system of a handful of particles the agreement is only approximate, and how good the approximation is becomes a question one can answer by direct computation. We diagonalize the two-dimensional hard-core Bose–Hubbard model completely—every eigenvalue in every particle-number sector—on square lattices of 4, 9, and 16 sites, and build all three ensembles from the one spectrum. Reading the temperature and chemical potential off the microcanonical entropy by finite differences turns out to be unreliable on spectra this small and this sharply bounded, so we track equivalence instead through quantities that are smooth sums over the spectrum: the relative energy fluctuation $\sigma_E/|\langle E\rangle|$ and the relative particle-number fluctuation $\sigma_N/\langle N\rangle$, both of which fall as $1/\sqrt{M}$ when the ensembles converge. For the free gas and for repulsion they do fall, with fits close to that rate. Attraction is different. The number fluctuation stays large, and for $V/t = -3$ its finite-size exponent is positive—the canonical-to-grand mismatch *grows* with the lattice rather than shrinking, which is what a lattice gas on the verge of phase separation should do. Sweeping $V/t$ from $-3$ to $+3$ turns this into a single monotonic curve in the number channel.

## 1 Introduction

Write down a box of gas three different ways—fixed energy, fixed temperature, fixed chemical potential—and thermodynamics says you get the same answer. The statement is exact only when the number of particles runs to infinity. Everything actually computed, measured in a trap, or simulated on a lattice has a finite number of particles, and then the three descriptions differ. The size of that difference, and the way it shrinks as the system grows, is the subject of this paper.

We work with the hard-core Bose–Hubbard model on a square lattice: bosons that hop between neighboring sites with amplitude $t$, repel or attract neighbors with energy $V$, and are forbidden from sharing a site. It is a quantum lattice gas in the plain sense of the phrase—a gas of quantum particles sitting on a lattice—and it is realized by cold bosonic atoms in an optical lattice in the strongly interacting limit. (It has nothing to do with the "quantum lattice-gas automaton," a cellular-automaton algorithm that shares the name.) Two features make it a good test bed. The local Hilbert space is two states per site, so the whole spectrum is reachable by exact diagonalization for small lattices, and the interaction can be dialed from repulsive through free to attractive without changing anything else.

The plan is ordinary but the order matters: build the Hamiltonian sector by sector, check it against anything that can be computed another way, diagonalize, and only then assemble ensembles and compare. One result of that discipline is negative and worth stating early. The textbook route to the microcanonical temperature—differentiate $S = \ln \Omega$ with respect to energy—does not survive contact with a spectrum of a few thousand discrete, bounded levels. We show why below and route around it.

## 2 Model

On an $L \times L$ lattice with $M = L^2$ sites,

$$\hat{H} = -t \sum_{\langle i,j \rangle} \left( \hat{b}_i^\dagger \hat{b}_j + \hat{b}_j^\dagger \hat{b}_i \right) + V \sum_{\langle i,j \rangle} \hat{n}_i \hat{n}_j, \qquad \hat{n}_i = \hat{b}_i^\dagger \hat{b}_i, \quad (1)$$

with the hard-core condition $\hat{n}_i \in \{0, 1\}$, $\hat{b}_i^2 = (\hat{b}_i^\dagger)^2 = 0$. The sum is over nearest-neighbor bonds, counted once. We set $k_B = 1$, measure energy in units of $t$, and use periodic boundaries. Because $\hat{H}$ commutes with $\hat{N} = \sum_i \hat{n}_i$, it is block diagonal in particle number; each block is diagonalized on its own and the eigenvalues feed all three ensembles. In the occupation basis the hard-core operators carry no sign between sites, so a hop contributes exactly $-t$ and every block is a real symmetric matrix.

A basis state is a string of occupations stored as an integer, bit $i$ holding $n_i$. The $N$-particle sector has dimension $\binom{M}{N}$, and the sectors add up to $2^M$. The canonical and grand-canonical sums run over the whole spectrum, not a few low-lying states, so we diagonalize each block densely rather than with a Lanczos method. The half-filled sector sets the cost: $\binom{16}{8} = 12\,870$ for $L = 4$, a few minutes on a workstation. We cache the spectra once.

## 3 The three ensembles

Everything below starts from one object: the set of eigenvalues $E_{N,\alpha}$, where $N$ labels the particle-number sector

and $\alpha$ the level within it. The three ensembles are three ways of weighting that set.

### 3.1 Microcanonical

An isolated system with $N$ particles and energy in a window $[E, E+\Delta E)$ visits each accessible level with equal probability. Counting them,

$$\Omega(E, N; \Delta E) = \#\{\alpha : E \le E_{N,\alpha} < E+\Delta E\}, \qquad S = \ln \Omega. \tag{2}$$

The first law without a volume term, $dE = T\,dS + \mu\,dN$, gives temperature and chemical potential as slopes of the entropy surface,

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_N, \qquad \mu = -T\left(\frac{\partial S}{\partial N}\right)_E. \tag{3}$$

Both slopes have to be taken as finite differences, since $E$ is discrete and $N$ is an integer. The window width $\Delta E$ is a modeling choice that does not go away: $\Omega$ is a histogram, and its bin size shows up in every quantity derived from it.

### 3.2 Canonical

Put the system in weak contact with a bath and hold the total energy fixed. The probability of the system sitting in level $\alpha$ is proportional to the number of bath states compatible with it, $\Omega_{\rm bath}(E_{\rm tot} - E_\alpha) = e^{S_{\rm bath}(E_{\rm tot}-E_\alpha)}$. The system energy is tiny against the bath, so expanding to first order,

$$S_{\rm bath}(E_{\rm tot} - E_\alpha) \simeq \text{const} - \beta E_\alpha, \qquad \beta = \frac{\partial S_{\rm bath}}{\partial E} = \frac{1}{T}, \tag{4}$$

and the Boltzmann weight follows:

$$p_\alpha = \frac{e^{-\beta E_{N,\alpha}}}{Z_N}, \qquad Z_N(\beta) = \sum_\alpha e^{-\beta E_{N,\alpha}}. \tag{5}$$

Mean energy and its fluctuation are derivatives of $\ln Z_N$:

$$\langle E\rangle_N = -\frac{\partial \ln Z_N}{\partial \beta} = \sum_\alpha E_\alpha p_\alpha, \tag{6}$$

$$\mathrm{Var}(E) = \frac{\partial^2 \ln Z_N}{\partial \beta^2} = \langle E^2\rangle - \langle E\rangle^2. \tag{7}$$

The heat capacity is fixed by the fluctuation, not by any separate measurement: differentiating $\langle E\rangle$ with respect to $T$ and using $d\beta/dT = -\beta^2$,

$$C_V = \frac{\partial \langle E\rangle}{\partial T} = \beta^2\,\mathrm{Var}(E). \tag{8}$$

The Gibbs entropy $-\sum_\alpha p_\alpha \ln p_\alpha$ collapses to $S = \beta\langle E\rangle + \ln Z_N$, and $F_N = -T\ln Z_N = \langle E\rangle - TS$.

### 3.3 Grand canonical

Let the system trade particles with the reservoir as well. The same first-order expansion, now in both energy and number, with $\partial S_{\rm res}/\partial N = -\beta\mu$, gives

$$p_{N,\alpha} = \frac{e^{-\beta(E_{N,\alpha}-\mu N)}}{\Xi}, \qquad \Xi(\beta,\mu) = \sum_N e^{\beta\mu N} Z_N(\beta). \tag{9}$$

The mean number and its fluctuation are $\mu$-derivatives of $\ln \Xi$,

$$\langle N\rangle = \frac{1}{\beta}\frac{\partial \ln \Xi}{\partial \mu} = \sum_{N,\alpha} N\, p_{N,\alpha}, \qquad \frac{\partial \langle N\rangle}{\partial \mu} = \beta\,\mathrm{Var}(N), \tag{10}$$

the second being the compressibility sum rule, derived the same way as Eq. (8). Since $\mathrm{Var}(N) > 0$, $\langle N\rangle(\mu)$ increases monotonically, a fact we lean on when matching ensembles. The mean energy is $\langle E\rangle = -\partial_\beta \ln \Xi + \mu\langle N\rangle$.

### 3.4 When they agree, and by how much

The canonical partition function is a Laplace transform of the microcanonical count. Writing the level density as $\rho_N(E) = \sum_\alpha \delta(E - E_\alpha) \approx e^{S(E)}$,

$$Z_N(\beta) = \int dE\, \rho_N(E)\, e^{-\beta E} = \int dE\, e^{\,S(E)-\beta E}. \tag{11}$$

For large $M$ the integral is dominated by the energy $E^*$ where $S'(E^*) = \beta$—which is exactly the microcanonical relation Eq. (3) identifying the two temperatures. Expanding around the saddle, the canonical energy distribution is Gaussian with width $\sigma_E^2 = -1/S''(E^*)$. Entropy and energy are both extensive, so $S'$ is intensive and $S'' \sim \mathcal{O}(1/M)$; hence $\sigma_E^2 \sim M$ and

$$\frac{\sigma_E}{|\langle E\rangle|} \sim \frac{\sqrt{M}}{M} = \frac{1}{\sqrt{M}}. \tag{12}$$

The same argument applied to $\Xi = \sum_N e^{\beta\mu N} Z_N$, now with a saddle in $N$, gives $\sigma_N/\langle N\rangle \sim 1/\sqrt{M}$. These two ratios are the small parameters of ensemble equivalence: when they are small the canonical energy and grand-canonical number distributions are sharp, and fixing $T$ or $\mu$ is the same as fixing $E$ or $N$. They are also pure spectral sums, with no differentiation, which is what makes them usable on a small discrete spectrum.

### 3.5 Why we do not read $T$ off the entropy

Equation (3) is the honest definition of the microcanonical temperature, and on a small spectrum it is close to useless. $S(E) = \ln \Omega$ is a jagged histogram; its slope swings wildly, diverges wherever the matched energy lands in a gap, and changes sign past the middle of the band, where the density of states peaks and the lattice gas enters its negative-temperature regime. Figure 5 shows the damage. We tried surface, Gaussian-broadened, and volume

(Gibbs) entropies; none rescues a single-energy thermometer at these sizes. So Eq. (3) stays in the paper as a diagnostic, with its $\Delta E$ dependence reported, and the quantitative claims rest on Eq. (12) and its number-channel twin.

To match the ensembles at a common operating point we fix $N$ and $\beta$, take the canonical $\langle E\rangle_N(\beta)$, and solve $\langle N\rangle(\beta,\mu^*) = N$ for the chemical potential. Monotonicity of $\langle N\rangle(\mu)$ makes the root unique; Brent's method finds it to $10^{-10}$. The energy per site compared across ensembles at that point, $|\langle E\rangle_{\text{grand}} - \langle E\rangle_{\text{can}}|/M$, is the third metric we track.

# 4 Validation

Each check below is a regression test in the code. *(i)* Every sector block is symmetric to machine precision. *(ii)* Block dimensions are $\binom{M}{N}$ and sum to $2^M$. *(iii)* A separately written full-space Hamiltonian is block diagonal in $N$, and the union of per-sector spectra equals its spectrum to $10^{-10}$—particle number is conserved. *(iv)* For $V = 0$ on $L = 3, 4$ the one-particle sector reproduces the tight-binding band $-2t[\cos k_x + \cos k_y]$ to $10^{-15}$; on the $2 \times 2$ plaquette it gives the four-site ring $\{-2t, 0, 0, 2t\}$. *(v)* $\mathrm{Tr}\,\hat{H}$ equals the summed interaction, since hopping is off diagonal. *(vi)* Ground-state energies from a sparse Lanczos solver agree with the dense spectra ($E_0/t = -10.347$ at $L = 3$, $-18.000$ at $L = 4$ for $V = 0$). *(vii)* $C_V = \beta^2\mathrm{Var}(E)$ and $\partial_\mu\langle N\rangle = \beta\mathrm{Var}(N)$ match numerical derivatives of the means to $10^{-5}$. *(viii)* The limits come out right: $S \to \ln(\text{ground degeneracy})$ and $C_V \to 0$ as $T \to 0$; $S \to \ln(\text{dim})$ and $\langle E\rangle \to$ the spectral mean as $T \to \infty$.

# 5 Results

## 5.1 Spectrum, heat capacity, filling

The spectrum is symmetric about zero and peaked in the middle (Fig. 1)—a bounded band with a large zero-energy degeneracy, both generic to bipartite hopping. The heat capacity is a single broad bump that vanishes at both ends of the temperature range (Fig. 2); repulsion pushes the bump to higher $T$ and attraction to lower, tracking the change in the characteristic energy scale. The grand-canonical filling passes through half-filling at $\mu = 0$ by particle–hole symmetry, and the $L = 3, 4$ curves have already nearly collapsed onto a common shape while $L = 2$ still stands apart (Fig. 3). Entropy per site climbs from zero at $T = 0$—a non-degenerate ground state—to the infinite-temperature ceiling $\ln(\text{dim})/M$, which itself creeps up toward $\ln 2$ as the lattice grows (Fig. 4).

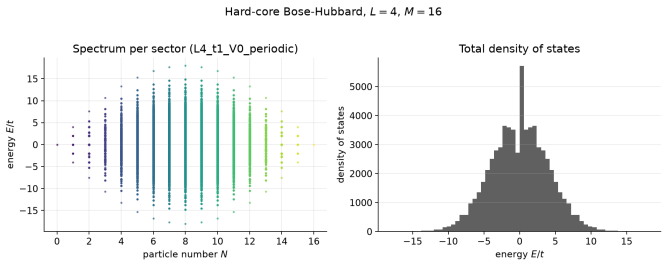



**Figure 1:** Eigenvalues per sector (left) and total density of states (right), $L = 4$, $V = 0$.

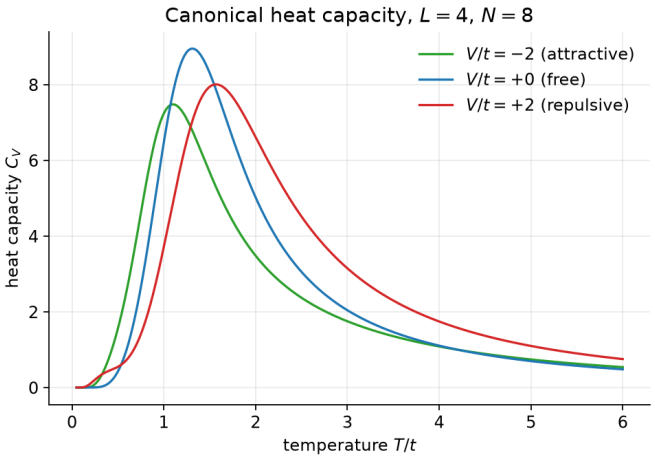



**Figure 2:** Canonical $C_V(T)$ at half filling ($L = 4$, $N = 8$).

## 5.2 The caloric curve, and the thermometry problem

Figure 5 is the clearest single picture of why the microcanonical temperature is not to be trusted here. The canonical $T(E)$, obtained by inverting the smooth $\langle E\rangle(T)$, rises cleanly through the lower half of the band. The finite-difference $1/(\partial S/\partial E)$ scatters around it below the band center and then goes negative above it. The negative values are not a bug—a bounded spectrum genuinely has negative-temperature states—but they make the slope useless as a thermometer at a single energy.

## 5.3 Matching the ensembles, and the finite-size trend

Fix $N$ and $\beta$, tune $\mu^*$ so $\langle N\rangle = N$, and the canonical and grand-canonical energies per site sit almost on top of each other, splitting by a small amount at intermediate temperature (Fig. 6). That gap, and the two fluctuation ratios, are what we follow against size.

Figure 7 and Table 1 collect the three metrics versus $M$ for seven couplings. For the free gas and for repulsion all three fall with size; the free-gas $\sigma_N/\langle N\rangle$ and per-site gap follow clean power laws ($R^2 = 1.00$) with exponents between $-0.6$ and $-0.9$, straddling the $M^{-1/2}$ that Eq. (12) predicts. The energy ratio $\sigma_E/|\langle E\rangle|$ is the noisiest of the

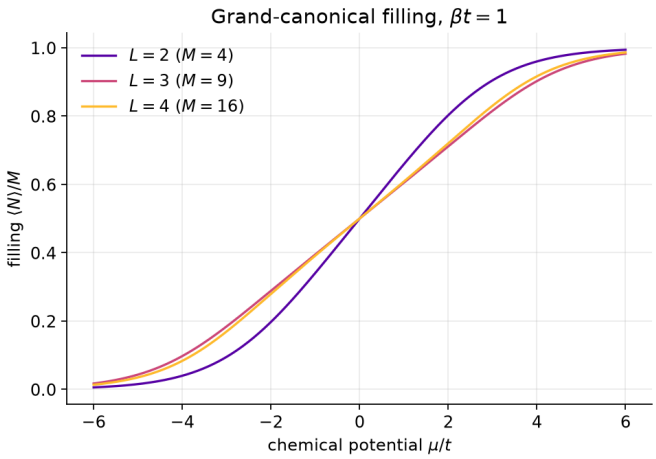



**Figure 3:** Grand-canonical filling versus $\mu/t$ at $\beta t = 1$.

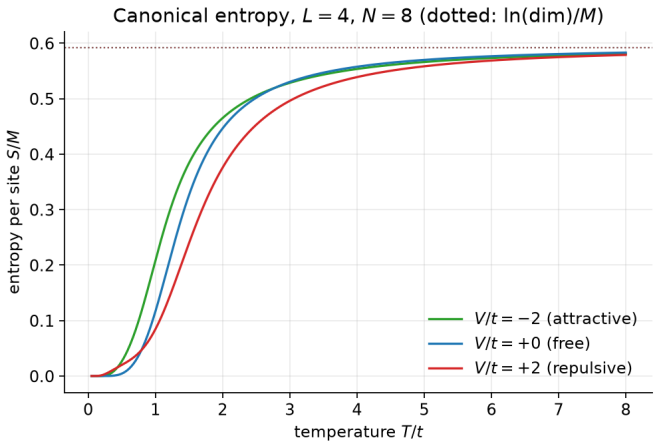



**Figure 4:** Entropy per site $S/M$; dotted lines mark $\ln(\dim)/M$.

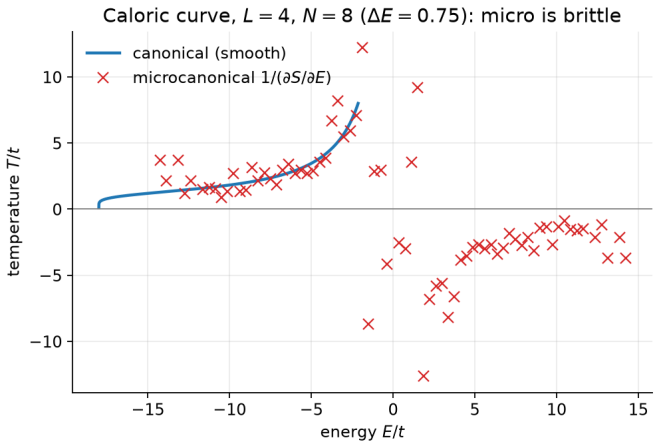



**Figure 5:** Caloric curve: smooth canonical $T(E)$ (line) against microcanonical finite differences (crosses), $L = 4$, $N = 8$, $\Delta E = 0.75$.

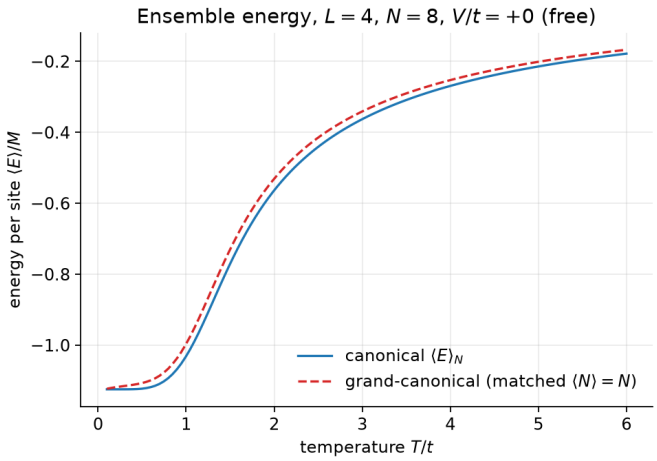



**Figure 6:** Canonical versus grand-canonical energy per site at matched $N$ ($L = 4$, $N = 8$, $V = 0$).

three, as expected—it feels the bounded band and the discreteness of the smallest lattice most strongly.

The attractive side behaves differently, and the difference is the main result. Reading down the $p(\sigma_N/\langle N \rangle)$ column of Table 1, the decay exponent slides monotonically from $-1.10$ at $V/t = +3$, where repulsion restores equivalence fastest, through $-0.64$ for the free gas, up to $+0.13$ at $V/t = -3$. The sign flip is the point. At strong attraction the relative number fluctuation reaches 0.96 on the largest lattice and its exponent is positive, so the canonical-to-grand mismatch *grows* as the lattice grows. An attractive lattice gas wants to clump; the grand ensemble lets the particle number swing between the competing dilute and dense arrangements, and the fixed-$N$ canonical ensemble cannot follow. The energy metrics keep shrinking for every $V$, so the failure lives specifically in the particle-number channel, which is where the instability lives too.

## 5.4 Interaction dependence

Plotting the metrics at $L = 4$ against $V/t$ lays the trend bare (Fig. 8): $\sigma_N/\langle N \rangle$ falls monotonically from nearly 1 at $V/t = -3$ to a few percent at $V/t = +3$, crossing the energy ratio near $V = 0$. A finer sweep at fixed $L = 3$, where each coupling is almost free to compute, resolves the same curve over 33 points and shows it is smooth (Fig. 9).

## 5.5 What to distrust

Three sizes make for soft exponents, and the smallest point does the most to tilt a fit. It also happens to be the oddest lattice: the $L = 2$ torus de-duplicates to a single-bonded four-site ring instead of the true double-bonded two-torus, with a one-particle band $\{-2t, 0, 0, 2t\}$ rather than the $-4t$ bandwidth minimum. The $L = 3$ torus has its own quirk—an odd period makes it non-bipartite, so it loses the $E \leftrightarrow -E$ spectral symmetry that $L = 2$ and

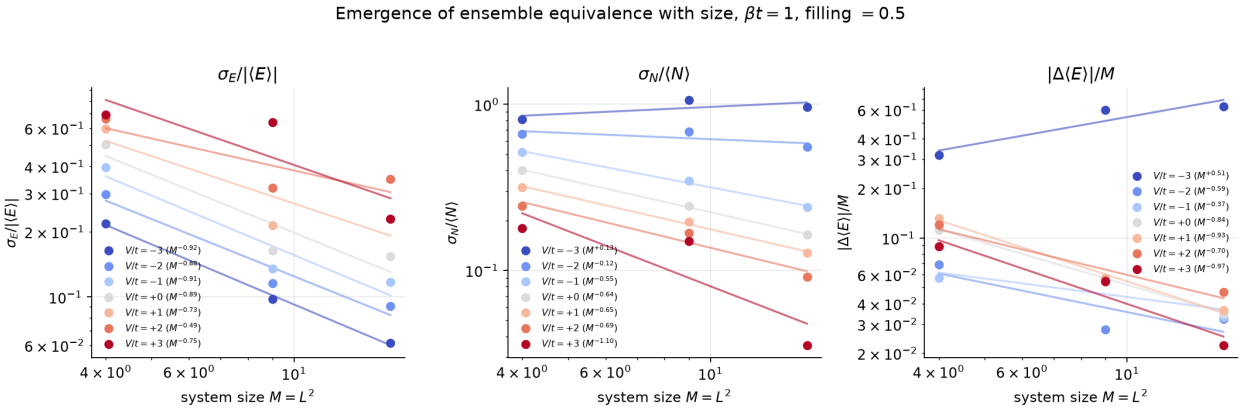



**Figure 7:** Finite-size scaling of $\sigma_E/|\langle E\rangle|$, $\sigma_N/\langle N\rangle$, and the per-site energy gap for seven couplings (attractive blue, repulsive red), $\beta t = 1$, half filling. Lines are power-law fits.

**Table 1:** Fitted exponents $p$ (metric $\propto M^p$, $M = 4, 9, 16$) and the number fluctuation on the largest lattice, $\beta t = 1$, half filling. The number-channel exponent rises steadily and turns positive at strong attraction.

| $V/t$ | $p(\sigma_E/\lvert\langle E\rangle\rvert)$ | $p(\sigma_N/\langle N\rangle)$ | $p(\Delta_E/M)$ | $\sigma_N/\langle N\rangle$ ($M$=16 |
|---|---|---|---|---|
| $-3$ | $-0.92$ | $+0.13$ | $+0.51$ | 0.96 |
| $-2$ | $-0.88$ | $-0.12$ | $-0.59$ | 0.55 |
| $-1$ | $-0.91$ | $-0.55$ | $-0.37$ | 0.24 |
| 0 | $-0.89$ | $-0.64$ | $-0.84$ | 0.16 |
| $+1$ | $-0.73$ | $-0.65$ | $-0.93$ | 0.12 |
| $+2$ | $-0.49$ | $-0.69$ | $-0.70$ | 0.09 |
| $+3$ | $-0.75$ | $-1.10$ | $-0.97$ | 0.05 |

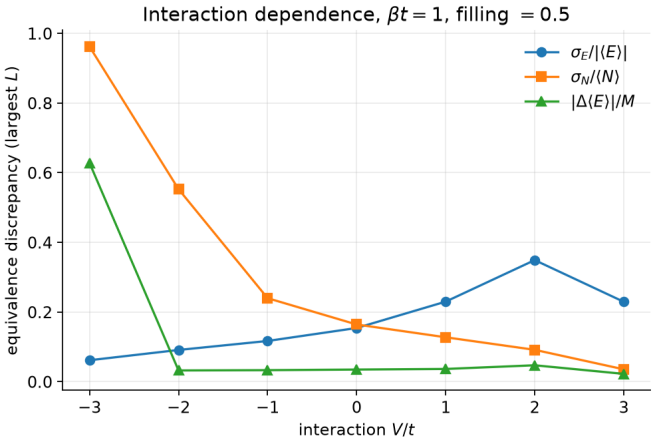



**Figure 8:** Equivalence metrics at $L = 4$ versus $V/t$, seven couplings.

$L = 4$ keep, though the $N \leftrightarrow M - N$ symmetry that pins $\langle N\rangle = M/2$ at $\mu = 0$ survives at every size. The $L = 2, 3, 4$ sequence is therefore three somewhat different geometries stitched together, and some of the scatter in Table 1 is that stitching rather than physics. An open-boundary sequence, bipartite at every size, or an even-$L$ series would be cleaner at the price of stronger edges. Everything here is at one temperature, $\beta t = 1$.

# 6 Discussion

The choice to measure equivalence through fluctuations rather than through a matched temperature is not a workaround; it is the content of the saddle-point argument in Sec. III D. Equivalence *is* the statement that the Laplace transform relating the ensembles is dominated by a sharp saddle, and the width of that saddle is exactly $\sigma_E/|\langle E\rangle|$. The quantity we can compute cleanly is the quantity that was physically at issue all along.

The attractive anomaly reads naturally as the finite-size shadow of a first-order transition. Phase coexistence is the standard place where ensemble equivalence breaks even in the thermodynamic limit, because the microcanonical entropy loses concavity and the Maxwell construction separates the ensembles. We do not have a transition on 16 sites, but we have its precursor: a number distribution broad enough that fixing $\mu$ and fixing $N$ stop being interchangeable, and broadening rather than sharpening as the lattice grows.

What would sharpen the picture is more sizes and a temperature axis. The exponents in Table 1 are suggestive, not converged, and the whole table lives at $\beta t = 1$; the rate at which equivalence sets in surely depends on temperature, and mapping that dependence is the obvious next step. Reaching $L = 5$ would need symmetry-reduced diagonalization, since the half-filled sector there is already five million states.

# 7 Conclusion

Built from one spectrum, the three ensembles of a small quantum lattice gas agree to a degree that improves with size for repulsive and non-interacting particles and fails to improve—indeed worsens—for strongly attractive ones, the failure confined to the particle-number channel where

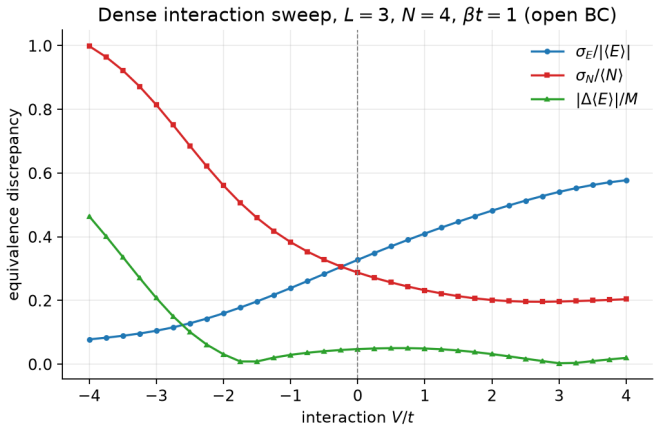



**Figure 9:** A denser sweep at fixed $L = 3$ (33 couplings, open BC, $\beta t = 1$): $\sigma_N/\langle N\rangle$ climbs smoothly toward unity with attraction.

the gas would rather phase separate. Getting there required recognizing that the microcanonical thermometer breaks on small bounded spectra and measuring equivalence through its fluctuations instead, which is both what works numerically and what the saddle-point picture says to measure.

**Author contributions.** P. Das, A. Shakya, and D. Thakur conceived the study, developed the exact-diagonalization and ensemble framework, performed the calculations and analysis, and wrote the manuscript.

**Reproducibility.** Each cached spectrum and figure records the generating configuration ($t$, $V$, $L$, boundary condition, grids, $\Delta E$). Spectra are built once and the analysis reads the cache; the validation checks of Sec. IV run as a test suite.

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